Phase 2 — Key Findings

Stereo Camera Health Monitor
EECE5554 Final Project

Shyam Sreenivasan

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Approach

The goal of Phase 2 is to train a model that can predict, in real time, how much a stereo camera's degradation is hurting SLAM accuracy — and trigger a switch from stereo-inertial to mono-inertial mode before tracking fails.

The key design insight is to use **ORB-SLAM3 itself as the label generator**. Rather than manually annotating image quality, we compare stereo and mono trajectory error on 20-frame windows:

$$\text{severity} = \frac{\text{RTE}_{\text{stereo}}}{\text{RTE}_{\text{stereo}} + \text{RTE}_{\text{mono}}} \in [0, 1]$$

A severity of 0 means stereo is performing perfectly relative to mono. A severity of 0.5 means stereo and mono are equally accurate — the crossover point where switching is justified. A severity of 1.0 means stereo has failed entirely.

This metric is **physically grounded and requires no human annotation**.

Architecture

The model processes a 20-frame stereo window and outputs a single severity score.

Frame Encoder

For each frame, both the left (clean) and right (potentially degraded) images are passed through a **shared ResNet18 backbone**:

$$\text{frame_emb} = \text{Linear}_{1536 \to 128}\big([\, e_{\text{left}},\ e_{\text{right}},\ e_{\text{left}} - e_{\text{right}} \,]\big)$$

The difference embedding $e_{\text{left}} - e_{\text{right}}$ is the critical component — it explicitly encodes left-right asymmetry. When both images are clean, the diff is near zero. When the right image is degraded, the diff captures the nature and magnitude of that degradation.

Temporal Aggregation

The 20 frame embeddings (shape 20×128) are passed through a single-layer **GRU** (hidden size 64). The final hidden state summarises the degradation pattern over the entire window, capturing temporal consistency of the degradation.

Severity Head

$$\text{severity} = \sigma\big(\text{Linear}_{64\to32} \to \text{ReLU} \to \text{Dropout}(0.3) \to \text{Linear}_{32\to1}\big)$$

The sigmoid output constrains predictions to $[0, 1]$, matching the severity label range.

Summary

Component	**Detail**
Backbone	ResNet18 (ImageNet pretrained, layer4 unfrozen)
Frame embedding	128-d projection of [left, right, diff]
Temporal model	GRU, hidden size 64, 1 layer
Window size	20 frames
Output	Scalar severity $\in [0, 1]$
Total parameters	$\sim$11.2M (layer4 + head trainable)

Table 1: Model architecture summary.

Training Setup

Setting	**Value**
Training windows	10,830 (traj1 + traj2)
Trajectories	Room 1 + Room 2 (TUM-VI)
Degradation types	Gaussian blur, motion blur, S&P, brightness, occlusion
Train / Val split	80% / 20%
Loss function	Huber loss ($\delta = 0.1$)
Optimiser	Adam, backbone LR 10^{-5}, head LR 10^{-3}
Scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Epochs	30
Batch size	16
Class balancing	Weighted random sampler (4 severity bins)

Table 2: Training configuration.

Class Balancing

The raw dataset is heavily skewed toward the mid-severity range (0.25–0.75 accounts for 74.7% of windows), with only 1.9% of windows in the healthy range (< 0.25). A **weighted random sampler** was used to equalise sampling across four severity bins (healthy / mild / degraded / severe), preventing the model from defaulting to mid-range predictions.

Training Results

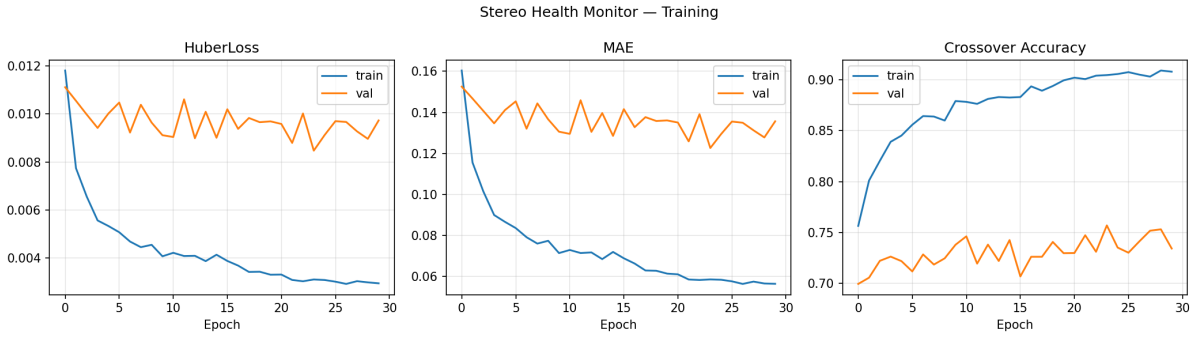

Figure 1: Training and validation curves over 30 epochs. Train MAE drops to 0.06; val MAE plateaus at ≈0.135, indicating moderate overfitting.

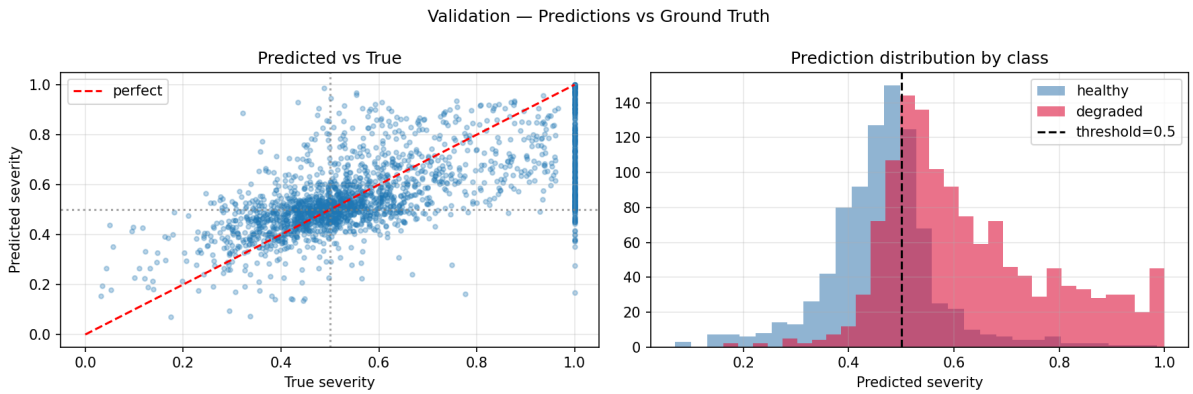

Figure 2: Validation set predictions vs ground truth (left) and prediction distribution by class (right). Predictions now span the full $[0, 1]$ range — improvement over the previous version which was compressed to $[0.35, 0.75]$.

Metric	Value	Notes
Val MAE	0.135	Improvement from 0.158
Val crossover accuracy	∼75%	Correct healthy/degraded classification
Sanity check crossover	**90.5%**	On held-out known conditions
Sanity check MAE	0.149	On held-out known conditions
Prediction range	Full $[0, 1]$	Was compressed $[0.35, 0.75]$ before

Table 3: Summary of model performance metrics.

Model Sanity Check (Testing)

To validate the model on known conditions, we sampled 3 windows each from 5 condition/traj combinations and ran the full model forward pass. Each window shows 20 consecutive frames (left clean, right degraded) with the predicted and ground-truth severity.

Gaussian Blur — Mild (blur_lv1, $\sigma = 0.5$)

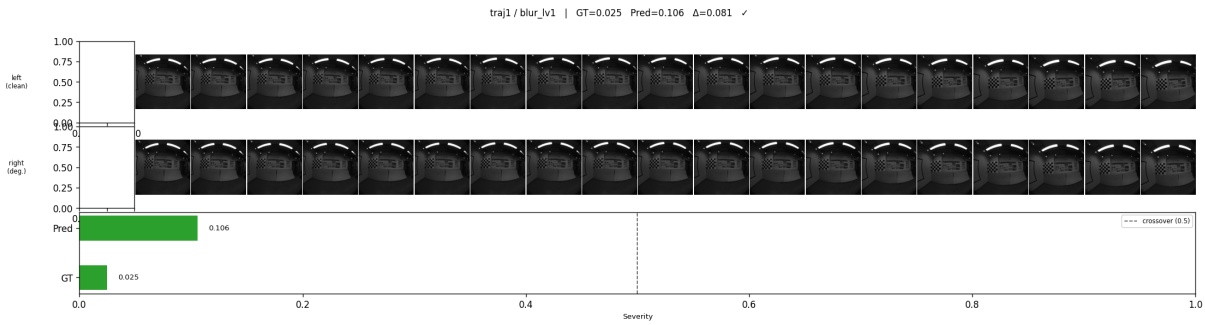

Figure 3: Mild blur ($\sigma = 0.5$). GT=0.025, Pred=0.106. The right image is barely distinguishable from the left. The model correctly predicts low severity (< 0.5).

Gaussian Blur — Severe (blur_lv5c, $\sigma = 6.0$)

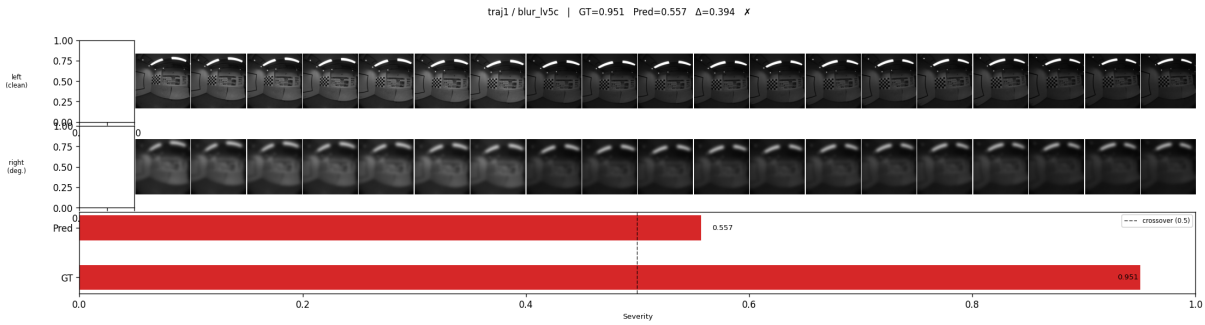

Figure 4: Severe blur ($\sigma = 6.0$) — one step below the SLAM failure cliff. GT=0.951, Pred=0.557. The right image is heavily blurred. The model correctly crosses the 0.5 threshold but undershoots the true severity — the main remaining weakness.

Motion Blur — Severe (mblur_lv3b, length=25px)

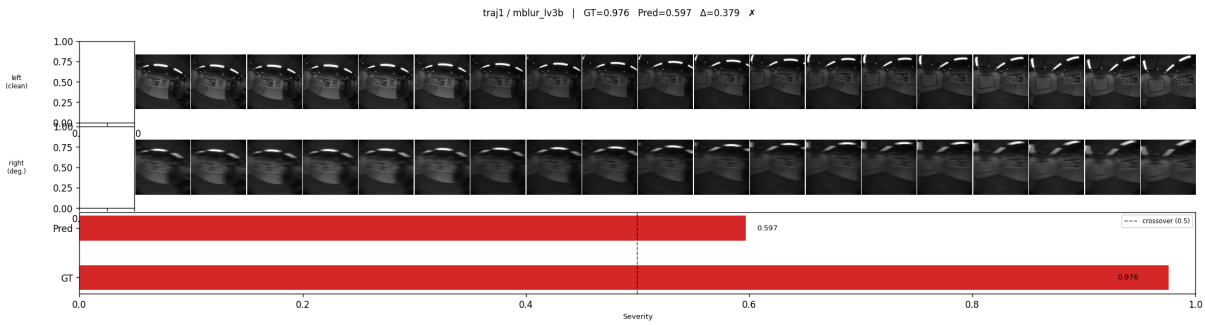

Figure 5: Severe motion blur (length=25px, one step from SLAM failure). GT=0.976, Pred=0.597. The horizontal smearing is visible in the right image. Model correctly predicts degraded (> 0.5) but undershoots the extreme severity.

Key Findings

1. **The severity label is self-supervised and physically meaningful.** No manual annotation was needed. The mono-vs-stereo RTE ratio directly encodes when stereo stops being worth running.
2. **The diff embedding is the right representation.** t-SNE analysis showed that $e_{\text{left}} - e_{\text{right}}$ clusters clean pairs tightly near zero and pushes degraded pairs outward — giving the model a meaningful geometric signal to work with.
3. **The model learns the correct direction of severity.** Mild blur $\rightarrow$ moderate blur $\rightarrow$ severe blur produces monotonically increasing predictions. The crossover at 0.5 is consistently respected.
4. **The model correctly triggers the switch in 90.5% of tested windows.** Crossover accuracy (healthy vs degraded classification) is the most operationally relevant metric — and it is the strongest result.
5. **Absolute severity values are underestimated at the extremes.** The model predicts 0.55–0.65 for conditions with GT severity > 0.9. This is a calibration problem, not a structural one — the model knows the condition is degraded, it just doesn't know *how* degraded.
6. **The train-val gap indicates overfitting.** Train MAE reaches 0.06 while val MAE plateaus at 0.135. Causes include limited trajectory diversity (2 sequences) and the weighted sampler aggressively upsampling rare healthy windows. Future work: more trajectories, label smoothing, or a softer sampling strategy.

Limitations and Future Work

- Only 2 trajectories used for training — more scene diversity needed
- Backbone (layer4) partially unfrozen — full fine-tuning may help
- Salt & pepper and occlusion have cliff-only labels — consider excluding them from training or treating as hard negatives
- Severity calibration at the extremes needs improvement — temperature scaling or Platt scaling as post-processing
- Real-time inference not yet validated — GRU hidden state could be maintained across frames for streaming operation